

Nita Kanney, MPH

Doctoral Candidate

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EDUCATION

University of Michigan (UM)

Candidate for Doctor of Philosophy (PhD) in Epidemiology, August 2023 – present
Committee: Kelly M. Bakulski (Chair), Belinda Needham, Devon Payne-Sturges, Margaret Hicken, Erin B. Ware

University of Alabama at Birmingham (UAB)

Master of Public Health (MPH) in Epidemiology, January 2021– December 2022
Advisor: Shakia Hardy

University of Maryland Baltimore County (UMBC)

Bachelor of Arts in Biological Sciences, minor in Critical Sexuality Studies, August 2015– May 2019
Advisor: Cynthia Wagner

PUBLICATIONS

Published

1. **Kanney N**, Patki A, Chandler-Laney P, Garvey WT, Hidalgo BA. (2022). Epigenetic Age Acceleration in Mothers and Offspring 4-10 Years after a Pregnancy Complicated by Gestational Diabetes and Obesity. *Metabolites*, 12(12), 1226. PMID: [36557264](https://pubmed.ncbi.nlm.nih.gov/36557264/).
2. Footman A, **Kanney N**, Niccolai LM, Zimet GD, Overton ET, Davies SL, Van Der Pol B. (2022). Perceived Need and Acceptance of a Future Chlamydia Vaccine among Healthcare Providers. *Sexually transmitted diseases*, 49(11), 740–744. PMID: [35709184](https://pubmed.ncbi.nlm.nih.gov/35709184/).
3. Footman A, **Kanney N**, Niccolai LM, Zimet GD, Overton ET, Davies SL, Van Der Pol B. (2022). Chlamydia Vaccination: Parent Opinions and Implications for Future Promotion Programs. *Sexually transmitted diseases*, 49(11), 745–749. PMID: [35948303](https://pubmed.ncbi.nlm.nih.gov/35948303/).
4. Footman A, **Kanney N**, Niccolai LM, Zimet GD, Overton ET, Davies SL, Van Der Pol B. (2022). Parents' Acceptance of COVID-19 Compared to Human Papillomavirus Vaccines. *Journal of adolescent health: official publication of the Society for Adolescent Medicine*, 71(6), 673–678. PMID: [36208985](https://pubmed.ncbi.nlm.nih.gov/36208985/).

Accepted

5. Li X, Juardo Velez J, **Kanney N**, Landrum A, Edwards LJ, Gutierrez OM, Long L, Hidalgo B. (2025). Incident Type 2 Diabetes in Individuals with Chronic Kidney Disease: The REGARDS Study. *Kidney Medicine*.

6. DeAngelis R, **Kanney N**, Hickey MT. Residential Contexts and Epigenetic Age Acceleration among Older-Age Black and White Americans: Findings from the Health and Retirement Study, 2010-2016. *Social Science & Medicine*.

Under Review

7. **Kanney N**, Cockell S, Wang H, Fu M, Dou, J, Hicken, MT, Payne-Sturges D, Needham BL, Ware EB, Bakulski KM. Early life experiences are associated with later life DNA methylation signatures in the Health and Retirement Study. Submitted to *Epigenetics*.

In Preparation

8. Fu M, **Kanney N**, Cockell S, Hornish U, Bakulski KM, Ware EB. Behavioral Risk Factors Associated with Cognition and Differences in DNA Methylation in the Health and Retirement Study. *In preparation*.
9. **Kanney N**, Rigby D, DeAngelis R, Dou J, Payne-Sturges D, Needham B, Ware EB, Bakulski KM, Hicken MT. Residential proximity to industrial pollution and later life differences in DNA methylation age acceleration in the Health and Retirement Study. *In preparation*.
10. **Kanney N**, Rigby D, DeAngelis R, Dou J, Payne-Sturges D, Needham B, Ware EB, Bakulski KM, Hicken MT. Epigenome-wide association study of residential proximity to industrial pollution in the Health and Retirement Study. *In preparation*.

RESEARCH EXPERIENCE

Graduate Student Research Assistant, UM, Dept. of Epidemiology August 2023–July 2025
Led original environmental epigenetics research in a population-based study. Attend weekly lab meetings for presentation of lab member projects, providing feedback on grants, and professional development. Meet with advisor to discuss research progress and academic updates. Use the Health and Retirement Study to produce an abstract analyzing early life exposures and DNA methylation aging in R statistical software for the International Society for Environmental Epidemiology (ISEE) conference. Also produced a manuscript and oral presentation for the American Public Health Association (APHA) conference.

Data Analyst, UAB, Dept. of Epidemiology January 2022–June 2023
Supported various research projects through data analysis of cohort data in R and SAS. Attend weekly meetings to discuss steps in the research plan. Conducted literature reviews in PubMed to draft manuscripts. Performed a Differentially Methylated Region analysis in RStudio using the dmrff package on epigenome-wide association study EPIC array data of gestational diabetes mellitus to evaluate the potential intergenerational transmission of gestational diabetes mellitus from mothers to offspring. Wrote three abstracts from research projects that resulted in presentations at a school-wide poster session with results from an epigenome-wide association study and two oral presentations at the APHA conference. Assisted as co-author in the preparation of a scoping review through data visualization of local studies in R statistical software and the extraction of data from manuscripts, which was accepted for a poster presentation at APHA. Used Ingenuity pathway analysis (IPA) and LocusZoom for assistance

with a meta-analysis of blood pressure and physical activity. Analyzed data on the American Heart Association Research Goes Red Precision Medicine Platform to conduct descriptive analysis in R statistical software and SAS on survey data of women's barriers and motivators for participation in research.

Summer Fellow, *National Institute of Child Health and Human Development*, June–July 2022
Conducted an epigenome-wide association study using the National Institutes of Health (NIH) Biowulf cluster with data from the Upstate KIDS study. Drafted a full-length manuscript of analysis results. Presented findings at study meetings and an institute-wide poster session. Attended health disparities intensive seminar series and omics journal club. Created science dissemination materials for a future study cohort and an infographic, and newsletters that communicated epigenetics to study participants from a broad audience.

Student Researcher, UAB, Department of Epidemiology June–October 2021
Analyzed and evaluated descriptive statistics on a longitudinal cohort data set in R statistical software. Discussed project updates with other researchers during bi-weekly genetic epidemiology lab meetings. Carried out literature searches in PubMed for a systematic literature review. Tracked selected manuscripts on EndNote Web and disseminated results with other researchers.

Research Assistant, UAB, Department of Health Behavior May 2021–April 2022
Communicated research project updates during weekly meetings with direct supervisor. Performed literature reviews and gave virtual monthly article presentations. Proofread transcripts, executed auto and manual coding of transcripts in NVivo using a thematic analysis approach. Assisted in the next steps of the qualitative analysis process by reviewing key themes and manuscripts preceding publication. Provided feedback on three manuscripts during the submission process.

TEACHING EXPERIENCE

Graduate Student Instructor, UM, Department of Epidemiology January–April 2026
● **EPID 668:** Professionalism, Practice, And Communication 2 (Instructor: Dr. Emily Martin)

Graduate Student Instructor, UM, Department of Epidemiology August–December 2025
● **EPID 670:** Integrated Problem Solving in Epidemiology 1 (Instructor: Dr. Abram Wagner)

Teaching Assistant, UAB, Department of Epidemiology August–December 2022
● **EPI 610:** Principles of Epidemiologic Research (Instructor: Dr. Stella Aslibekyan)

Teaching Assistant, UAB, Department of Epidemiology March–May 2022
● **PUH 603:** Quantitative Methods in Public Health (Instructor: Dr. Stella Aslibekyan)

PROFESSIONAL EXPERIENCE

Nutrition Research Associate, Mathematica

February–March 2020

Completed training on documenting discrepancies, using the Nutrient database, and communicating professionally with school nutrition managers. Reviewed nutrition menus for assigned set of schools by proofreading, and documenting incongruities with the established codebook in Excel. Updated the Nutrient database after supervisors approved the Excel document. Contacted school nutrition managers through email and phone for follow-up and data retrieval when necessary. Participated in regular meetings with supervisors to garner feedback, discuss personal and team performance, and provide suggestions on the data entry process.

Microbiology Analyst, EMSL Analytical

June–January 2020

Conducted daily Quality Control (QC) analyses and followed safety operating procedures when performing client analyses. Participated in monthly laboratory training. Performed appropriate laboratory tests and procedures within tight deadlines and performed laboratory cleaning. Used research databases to remain current on environmental practices. Carefully and consistently analyzed client samples, collaborating with lab managers and another analyst for specimen QC.

AWARDS & GRANTS

NIH T32 Interdisciplinary Research Training in Health and Aging (Fall 2026- Spring 2027)

Epidemiology Doctoral Student Organization (EDSO) School of Public Health Student Organization of the Year, UM (2025)

Patricia A. Peyser Endowed Scholarship, Department of Epidemiology, UM (2024)

Ian and Millicent Higgins Scholarship, Department of Epidemiology, UM (2023)

Rackham Conference Travel Grant, UM (2023–2025)

Delta Omega Honorary Society in Public Health, UAB Upsilon Chapter (2023)

School of Public Health Poster Day Award, UAB (2022)

PRESENTATIONS

1. **Kanney N**, Rigby D, DeAngelis R, Payne-Sturges D, Needham B, Ware EB, Bakulski KM, Hicken MT. Is residential proximity to industrial pollution associated with later life differences in DNA methylation age acceleration in the U.S. Health and Retirement Study? International Society for Environmental Epidemiology North America Conference (ISEE-NAC) 2026 conference in Baltimore, MD. *Poster presentation*.
2. **Kanney N**, Rigby D, DeAngelis R, Payne-Sturges D, Needham B, Ware EB, Bakulski KM, Hicken MT. Is residential proximity to industrial pollution associated with later life differences in DNA methylation age acceleration in the U.S. Health and Retirement Study? University of Michigan Aging Initiative Symposium: Stress, Resiliency, and Health in Later Life 2026 Symposium in Ann Arbor, MI. *Poster presentation*.
3. **Kanney NM**, Cockell S, Wang H, Ware EB, Bakulski KM. The embodiment of early life substance use and later life DNA methylation age acceleration in the Health and Retirement Study. American Public Health Association (APHA) 2025 annual meeting in Washington, DC. *Oral presentation*.

4. **Kanney NM**, Cockell S, Wang H, Ware EB, Bakulski KM. Childhood early life environment associated with GrimAge DNA methylation acceleration in the Health and Retirement Study. International Society for Environmental Epidemiology (ISEE) 2024 conference in Santiago, Chile. *Oral presentation*.
5. **Kanney N**, Hidalgo B. Evaluating causality of known risk factors of gestational diabetes mellitus: A two-sample mendelian randomization study. American Public Health Association (APHA) 2023 annual meeting in Atlanta, GA. *Oral presentation*.
6. **Kanney N**, Patki A, Chandler-Laney P, Garvey WT, Hidalgo B. Identifying differentially methylated regions in mothers and offspring 4-10 years after a gestational diabetes mellitus pregnancy: The hapi study. American Public Health Association (APHA) 2023 annual meeting in Atlanta, GA. *Oral presentation*.
7. Juado Velez J (presenter), Barron I, **Kanney N**, Nekayla N, Hidalgo B. (2023). Examining The Inclusion and Analysis of Race-Sex Groups in Primary Hypertension Studies Conducted in the US: A Comprehensive Review. American Public Health Association (APHA) 2023 annual meeting in Atlanta, GA. *Poster presentation*.
8. **Kanney N**, Polinski KJ, Yeung EH. Epigenome-wide association study of newborn dried blood spot DNA methylation with birthweight and gestational age. 2022 NIH Summer Research Program Poster Day. Virtual. *Poster presentation*.
9. Jurado Velez J (presenter), **Kanney N**, Shrestha S, Hidalgo B. (2022). Post-Acute Sequelae of SARS-CoV-2 and mortality among individuals with breakthrough infections in Alabama. American Heart Association (AHA) 2022 annual meeting in Chicago, IL. *Circulation*, 146(Supplement 1). https://doi.org/10.1161/circ.146.suppl_1.14914. *Poster presentation*.
10. **Kanney N**, Patki A, Chandler-Laney P, Garvey WT, Hidalgo B. (2022). Epigenetic aging 4–10 years after a gestational diabetes mellitus pregnancy. Virtual. *Annals of Epidemiology*, 75, 79. <https://doi.org/10.1016/j.annepidem.2022.08.035>. *Poster presentation*.
11. **Kanney N**, Patki A, Chandler-Laney P, Garvey WT, Hidalgo B. Epigenome-wide association study of pregnancies complicated by gestational diabetes and obesity: The HAPI Study. 2022 Public Health Research Poster Session at UAB School of Public Health in Birmingham, Al. *Poster presentation*.

PERSONAL PROJECTS

GitHub account: [NMK2022](https://github.com/NMK2022)

- **Data Management and Reproducible Code**, GitHub: https://github.com/NMK2022/EPID668_WK11_Titanic. Repository created for lecture presentation in UM EPID 668 on data management and reproducible code best practices.

Assist students with setting up and formatting a GitHub repository for an internship coding project in RStudio and example files for reproducible code using the base R titanic data set.

- **COVID-19 Analysis**, GitHub: <https://github.com/NMK2022/COVID19Analysis>. Created in response to the coronavirus pandemic. Using RStudio to visualize the per capita death rate globally over a five-month span.
- **Harvard-CS50x**, GitHub: <https://github.com/NMK2022/Harvard-cs50x>. A selection of projects I created during the course. Showcasing proficient ability to code in HTML and CSS. Further, a basic coding proficiency in Python and SQL
- **MD Infant Mortality**, GitHub: <https://github.com/NMK2022/MDMortalityAnalysis>. Infant mortality is a prevalent issue in rural areas and areas with a high population of African Americans. Using RStudio and public data from the Maryland Department of Health to visualize the disparities in infant mortality rates for seven Maryland counties.

LEADERSHIP & SERVICE

Future Public Health Leaders Program (Panelist, UM)

- July 6, 2026
- Panel Session: Tips and Strategies for Doctoral Programs Panel
- Presented my past and current experience to roughly 30 undergraduate students taking part in the Future Public Health Leaders Program. Alongside 3 other panelists, I answered questions about my path to a doctoral program and strategies used before and after my matriculation into the program. Engaged in open Q&A session with students to provide additional information on how they can be successful candidates for graduate education.

Complexity in the Social World (Session Co-discussant, UM)

- October 9, 2025
- Panel Session: Properties of structural racism as a framework to guide research
- Used my interdisciplinary experience and training, I presented a 5-minute introductory speech to a diverse audience on reframing the operationalization of traditional quantitative measures for studying structural racism to a theory-driven application. Co-moderated an open Q&A session among panelists and audience members, including facilitation of questions for attendee engagement.

Epidemiology Doctoral Student Organization (Social Co-Chair, UM)

- Fall 2024–Winter 2025
- Attend bi-weekly meetings through Zoom with other committee members. Worked with the other co-chair and the President to brainstorm and plan social events for doctoral students in the department. Awarded the UM School of Public Health best student organization of the year award in May 2025 at the Public Health Honors.

Women of Color and the Academy (Member, UM)

- Fall 2023–present
- Participate in virtual writing sessions to support degree progress, attend professional development sessions related to research writing, and provide support for other students. Participated in a mentorship event co-sponsored with the Black Psychology Student Association and Our Community Reads community organization. As graduate students, we connected with Black female students (4th to 9th grade) from the organization to expose them to the University of Michigan Art Museum, hold conversations about a new exhibit featuring 100 years of art from the Spelman College collection, share our experiences getting to graduate school, provide advice, and learn about the students' goals for the future.

Black Graduate Student Association (Treasurer and Member, UM)

- Fall 2023–present
- As a member, I participated in social events to network with other graduate students and faculty across the University. Volunteer alongside other members to plan the annual Black Graduate Research Symposium committee. Attended meetings to plan symposium conceptualization and events.
- In the treasurer role, I provide financial management and organizational support to the association to help with funding social, research, and networking events throughout the semester. Identified potential funding sources, leading to two successful student organization grant awards during the Winter 2026 semester through the Multi-Ethnic Student Affairs (MESA) department.

Society for Epidemiologic Research Mentorship Program (Member, SER)

- Fall 2022–Spring 2023
- Selected to participate in a 6-month mentoring program focused on career and professional development as a person of color. Learn about effective strategies to navigate mentor relationships and professional development.

Maternal and Child Health Association (President and Member, UAB)

- Member (Spring 2021–Fall 2022) and President (Fall 2021–Fall 2022)
- Participated regularly in monthly meetings, virtual seminars, and events. Served as President, where I led monthly meetings and helped coordinate events, like helping set up a book donation drive for United Way of Central Alabama.

Saint Agnes Hospital (Oncology Volunteer, UMBC)

- October–December 2017
- Volunteered through UMBC Shriver Center in the Oncology department. Observed nurses, doctors, and medical students on their rounds. Aided in patient intake and discharges. Assisted patients and nurses with daily tasks.

American Red Cross Club (Member and Volunteer, UMBC)

- Fall 2016–Spring 2019
- Attended biweekly club meetings to stay informed about club events and to participate in writing letters to veterans and making goodie bags for Ronald McDonald House children. Volunteered at the Ronald McDonald House in Baltimore, making crafts and interacting with kids and their parents. Assisted in running an on-campus blood drive by checking in donors, providing them with information, and answering their questions.

MEMBERSHIPS & AFFILIATIONS

Landscapes of Population Health	2025–present
International Society for Environmental Epidemiology (ISEE)	2024–present
Delta Omega Honorary Society in Public Health	2023–present
American Public Health Association (APHA)	2023–2026
Omicron Delta Kappa (ODK)	2022–present
Society for Epidemiologic Research (SER)	2022–present
Maryland Public Health Association (MdPHA)	2021–2022

RESEARCH SKILLS

Qualitative: Literature Review, Database Searches, Transcription, NVivo, Thematic Coding, Oral Presentations

Quantitative: R, RStudio, NIH Biowulf, Great Lakes HPC, SAS, JMP, HTML/CSS, Microsoft Word and Excel

CERTIFICATIONS

Responsible Conduct of Research and Scholarship (RCRS) Training, UM

- Issued September 11, 2023, Expires September 10, 2026

Program for the Education and Evaluation of Responsible Research and Scholarship (PEERRS) Training, UM

- Issued September 22, 2023, Expires September 21, 2026

R Programming, Johns Hopkins University on Coursera

- Credential ID: 488542UEG83H
- Issued April 13, 2020, No Expiration Date

CS50x, Harvardx on edx

- Credential URL <https://cs50.harvard.edu/certificates/408e8db8-f15c-4146-aa39-c3b008339ad5>
- Issued July 2020, No Expiration Date

HIPAA Privacy and Security Training, UAB

- Credential ID: C60667

- Issued December 2020, No Expiration Date

TRAININGS

Columbia Epigenetics Bootcamp

July 29–July 30, 2024

Engaged in two full-day intensive sessions on advanced epigenetic methodologies at the Columbia Mailman School of Public Health campus. Learned from distinguished researchers in seminar lectures, networked with other attendees, and conducted hands-on coding lab sessions using Posit Cloud. During lab sessions, the provided code was used to practice conducting an epigenome-wide association study. Including running code to clean the DNA methylation data through quality control and preprocessing steps, data visualizations, and global, regional, and site-specific methylation analysis.

University of Michigan Summer Session in Epidemiology

July 8–July 12, 2024

Participated in a virtual one week course on applied mediation analysis. Watched recorded lectures, attended live Zoom sessions, and completed practice assignments. Topics covered included practice applying a potential outcomes framework for causal inference, assumptions for mediation analysis, approaches for conducting regression analysis, how to account for unmeasured confounding, decomposing mediating and interactive effects, and approaches for handling multiple mediators.

RELEVANT COURSEWORK

- Applied Calculus
- Statistics with Applications in Biological Sciences
- Intermediate Statistical Analysis
- Intro to Data Analysis with SAS
- Qualitative and Mixed Methods Research in Public Health
- Critical Appraisal of Epidemiologic Studies
- Social Epidemiology II: Social and Economic Determinants of Population Health
- Analyzing Longitudinal and Clustered Data Using Statistical Software
- Community Assessments in Public Health
- Methodologies in Gender, Women's and Sexuality Studies
- Sex and Reproduction in the US
- Human Sexuality and the Sociological Perspective
- Sexuality and Queer Theory